bhttp/1: HTTP in binary frames

Version 1 · Raj Prakash · HTTP semantics in length-prefixed binary frames over one persistent TCP connection.

1. Conventions

MUST, MUST NOT, SHOULD, SHOULD NOT and MAY are used as in RFC 2119. Integers are unsigned and **big-endian** (network byte order); widths are in bits. A *message* is a request or a response.

2. Connection

bhttp/1 runs over TCP, with no registered port (examples use 9000) and no preface: the client's first bytes are its first frame.

- **Persistent and sequential.** Any number of requests, one at a time: a client MUST NOT send a REQUEST before the previous response's END has arrived; a server MUST answer in the order received.
- **The client closes** when it is done (the normal end); it MAY send GOAWAY first.
- **The server closes** (a) after a connection error (§7); (b) when idle: no complete REQUEST within T seconds of accepting the connection or finishing the previous response (inside a request body: of the body's last DATA frame); other frames and partial frames do not extend T, which SHOULD be at least 10 s; (c) to refuse a connection it cannot take, with GOAWAY Last-ID 0, Code `0x02` BUSY; the client MAY retry later.
- A server closing on its own initiative SHOULD send GOAWAY, then shut down its sending side and read and discard input for up to 2 s, so the peer reads the GOAWAY rather than a reset.
- A receiver whose connection ends mid-frame discards the partial frame and closes, sending nothing.

3. Frame header

Every frame is an 8-byte header followed by Length bytes of payload.

```
byte  0     1     2     3     4     5     6     7
    +-----+-----+-----+-----+-----+-----+-----+-----+
    |  Length   |Type |Flags|      Request ID       |
    +-----+-----+-----+-----+-----+-----+-----+-----+
         16        8     8            32
```

- **Length (16):** payload bytes after the header. MUST be ≤ **16384**; a larger value is a connection error (§7) whatever the Type, so a receiver never needs a buffer larger than 8 + 16384 bytes.
- **Type (8):** the frame type (§4).
- **Flags (8):** `0x01` = **END**. Other bits MUST be 0 when sent and MUST be ignored on receipt.
- **Request ID (32):** the request a frame belongs to. A client MUST number its requests 1, 2, 3, … on each connection, each exactly one more than the last; after 0xffffffff it MUST open a new connection. The server copies the ID into every frame of the response and checks only that it is not 0. ID 0 names the connection, as stream 0 does in HTTP/2: a REQUEST, RESPONSE or DATA with ID 0 sent to a server is a connection error (§7), and a RESPONSE with ID 0 reports one.

Why these widths. HTTP/2 (RFC 9113) has a 9-byte header: Length 24, Type 8, Flags 8, then 1 reserved bit and a 31-bit stream ID. Its reasons, and where bhttp/1 differs:

Field	HTTP/2 chose … because	bhttp/1 chooses … because
Length	**24.** Frames are limited by `SETTINGS_MAX_FRAME_SIZE`: 2^14 by default, up to 2^24 – 1 if the peer allows. The default is small because streams share the connection: one huge DATA frame would stall every other stream behind it (head-of-line blocking).	**16,** capped at the same 2^14. With no SETTINGS to raise it, a wider field would only describe frames nobody may send; the 16th bit lets a later version negotiate 64 KiB, and a small cap keeps a multiplexed v2 fair, as in HTTP/2. Overhead: 0.05 % of a full frame. Text such as `GET /` reads as Length ≥ 16640 and fails on its first two bytes.
Type	**8:** 256 types, 10 defined; unknown types MUST be ignored and discarded, which is how HTTP/2 is extended.	**8,** with the same rule (§4): it is what leaves room for version 2. v1 uses 4 types.
Flags	**8** per-type booleans (END_STREAM, END_HEADERS, PADDED, PRIORITY, ACK); one bit means different things in different types.	**8.** v1 needs one bit, END, with one meaning; seven stay free.
ID	**1 + 31.** R MUST be 0 when sent and is ignored on receipt. Client streams are odd, server-push streams even; IDs are never reused, so after 2^31 streams the connection must be replaced. Stream 0 is the connection itself.	**32,** no R bit, no odd/even split: without push or multiplexing, order alone matches responses. The ID stays so a client spots a desynchronised connection, and because the header is the one thing no later version may change (receivers must find Length in the same place to skip unknown frames): room for multiplexing is reserved now. 2^32 – 1 requests per connection; every field naturally aligned.

4. Frame types

Type	Name	From	ID	Payload
`0x01`	REQUEST	client	new	Method, Path, headers (§5.1)
`0x02`	RESPONSE	server	copied	Status, headers (§5.2)
`0x03`	DATA	both	same	0–16384 body bytes
`0x04`	GOAWAY	both	0	Last-ID (32), Code (8)

END on a REQUEST or RESPONSE means "no body follows"; on DATA, "last frame of the body"; on GOAWAY it is ignored.

Unknown types: the MUST-skip rule. A receiver that meets a frame whose Type it does not know MUST read and discard exactly Length payload bytes and carry on as if the frame were absent, whatever its Flags and ID and wherever it appears, also between the frames of a message. It MUST NOT reply to it or close because of it; only the Length limit applies. Types `0x00` and `0x05–0x7f` are reserved for later versions of this spec and `0x80–0xff` for private experiments; a v1 sender MUST NOT send them.

GOAWAY means "I am closing; send no more requests". Its ID is 0 and is ignored on receipt. *Last-ID*: from a server, the last request it answered completely (0 if none); a client sends 0. *Code*: `0x00` NO_ERROR (idle, shutdown), `0x01` PROTOCOL_ERROR, `0x02` BUSY (§2); any other value counts as `0x01`. Bytes after the first five are ignored; a shorter payload reads as Last-ID 0, Code `0x01`. The receiver MUST NOT send another REQUEST and SHOULD close; requests with an ID above Last-ID were not processed and MAY be retried on a new connection.

5. Payloads

5.1 REQUEST: `Method (8) | Path Length (16) | Path | header block`. Methods: `0x01` GET, `0x02` HEAD, `0x03` POST, `0x04` PUT, `0x05` DELETE, with HTTP meaning; any other code is malformed. The Path is 1 to Length – 3 raw bytes, never percent-decoded (`%2e%2e` is a literal name); everything from its first `?` is the query. The Path is malformed if it contains a byte `0x00–0x1f` or `0x7f`, or if the part before the query does not start with `/` or has a `/`-separated segment equal to `.` or `..`.

5.2 RESPONSE: `Status (16) | header block`, an HTTP status from 200 to 599 (v1 has no 1xx responses).

5.3 Header block. Zero or more entries, back to back, filling the rest of the payload exactly. Each starts with an Index byte:

```
indexed:  | Index 1..10 | VLen (16) | Value |
literal:  | 0x00 | NLen (8) | Name | VLen (16) | Value |
```

Index	Name	Index	Name
1	`host`	6	`last-modified`
2	`user-agent`	7	`date`
3	`accept`	8	`server`
4	`content-type`	9	`location`
5	`content-length`	10	`allow`

These are exactly the ten names bcurl and bserve send (1–3 client, 4–10 server). A name in the table SHOULD be sent indexed; receivers MUST accept the literal form too. An entry is malformed if its Index is 11–255, if it runs past the payload, if a literal Name is empty or has a byte other than lowercase `a-z 0-9 ! # $ % & ' * + - . ^ _ | ~` and backtick, or if a Value contains `0x00`, `0x0a` or `0x0d`. Order is kept and repeated names have HTTP meaning; `date` and `last-modified` use IMF-fixdate (`Tue, 13 Oct 2026 05:52:22 GMT`). Clients SHOULD send `host`; a server MUST NOT require any header. The block must fit in its one frame.

Payload widths. A byte suffices for Method, Index and NLen (5 methods, 10 names, names ≤ 255 bytes); Path Length, VLen and Status take 16 bits (long paths and values; 599 > 255). As in HPACK the static table is 1-based and Index 0 means "a literal name follows"; unlike HPACK, integers are fixed-width (no prefix-varints) and there is no Huffman coding or dynamic table, which save bytes but need a stateful decoder.

6. Messages and bodies

A message is a REQUEST or RESPONSE frame and, if it lacks END, a body of DATA frames with the same ID, the last carrying END. Only a REQUEST opens a body at a server. Senders SHOULD fill every DATA frame but the last; an empty one is allowed (useful with END).

`content-length` is optional. If sent, its value is 1 to 19 ASCII digits (leading zeros allowed), any further copy MUST be byte-for-byte identical (`3` and `03` are not), and it MUST equal the body length. The exception is HEAD: every response to a REQUEST whose Method byte is HEAD, whatever its status (a 400 too), sets END on the RESPONSE and has no DATA; a 200 carries the headers a GET would get. A receiver treats a violation as malformed.

A server MUST read a request body through its END before it treats a further frame as a new request. A server that fails after sending RESPONSE (say, a read error mid-file) MUST close without sending END.

7. Errors

Connection errors break the framing; the stream cannot be resynchronised: a Length above 16384 (the frame is not read further); a REQUEST, RESPONSE or DATA with ID 0; and, while a request body is open, any *known* frame type other than DATA for that ID (unknown types are still skipped, §4; a GOAWAY just ends the connection, with no 400). The server sends RESPONSE **400** with ID 0 (optionally with a body under that ID), then GOAWAY `0x01`, and closes as in §2.

Request errors leave the framing intact: the next frame starts where expected. The server answers RESPONSE **400** with the offending frame's ID and **keeps the connection open**. They are a REQUEST payload under 4 bytes, an unknown method, a Path Length of 0 or past the payload, a Path rule violation (§5.1), a malformed header block, a bad `content-length` (§6), a RESPONSE sent to a server (it opens no body), and a DATA frame with no open request body. A malformed REQUEST without END still has its body read first.

Other replies: 404 not found, outside the root, hidden, or not a regular file; **403** not readable; **301** with `location` = the path without its query, plus `/`, for a directory named without a trailing `/`; **405** with `allow` for a method the server does not implement; **500** for a read error before RESPONSE. Error responses MAY carry a short `text/plain` body.

Clients: a frame with Length > 16384, a RESPONSE or DATA with a wrong ID (a RESPONSE with ID 0 reports a connection error), DATA before the RESPONSE, a second RESPONSE, a REQUEST, a malformed payload, a status outside 200–599, a bad `content-length`, or EOF before END is a protocol error: close and report failure. A GOAWAY before END means the request was not processed: close; it MAY be retried on a new connection.

8. Path mapping

A server serving the directory ROOT drops the query; appends `index.html` if the path ends in `/`; answers 404 if a segment starts with `.`; resolves ROOT + path following symbolic links and answers 404 if the result is outside ROOT (resolved the same way); answers 301 (§7) for a directory, or 404 if the path already ended in `/`; and 404 for anything else that is not a regular file. This MUST hold for what is actually opened even if the tree changes meanwhile (say, open the resolved name component by component from ROOT without following links, so a link swapped in yields 404). Syntax errors are 400 (§5.1), so `..`, a Path without its leading `/` and NUL bytes never reach the file system. `content-type` SHOULD follow the file extension (`application/octet-stream` if unknown).

9. Versions

No version number is sent; the frame header and the MUST-skip rule are frozen for every version. A later version may add frame types (`0x05-0x7f`), flag bits, GOAWAY codes and bytes, and frames up to Length 65535, but uses them only once the peer has shown it understands them: a v2 peer announces itself with a new frame type (say, HELLO) that a v1 peer silently skips. A v1 server never sends one, so a v2 client that gets a RESPONSE without it stays at v1.

10. Worked example

GET `/` with no headers, ID 1, answered by a 404 with no body:

```
00 04 01 01 00 00 00 01   Length 4, REQUEST, END, ID 1
01 00 01 2f               GET, Path Length 1, "/"
00 02 02 01 00 00 00 01   Length 2, RESPONSE, END, ID 1
01 94                     Status 0x0194 = 404
```

`01 00 0e localhost:9000` is `host: localhost:9000`; `00 03 x-a 00 01 1` is the literal `x-a: 1`. HEXDUMP.md annotates a full exchange. One connection, frame by frame:

```
C>S  REQUEST  id=1 END   GET /a.bin
S>C  RESPONSE id=1       200, content-length: 20000
S>C  DATA     id=1       16384 bytes
S>C  DATA     id=1 END   3616 bytes
C>S  type 0x42 id=0      unknown: skipped, no reply
C>S  REQUEST  id=2 END   GET /x/../y
S>C  RESPONSE id=2       400: request error, kept open
S>C  DATA     id=2 END   "400 Bad Request: ...\n"
       ... T seconds without a REQUEST ...
S>C  GOAWAY   id=0       Last-ID 2, NO_ERROR; S closes
```

11. Reference client (informative)

`bcurl [-v] [-t seconds] URL...` with URL = `[bhttp://]host[:port][path]`: host a name, IPv4 address or `[IPv6]`, port 9000 by default. The host ends at the first `:` `/` `?` or `#`; a `#fragment` is not sent; an empty path is `/`, and `?q` is `/?q`. All URLs share one host and port and go in order over one connection, never reopened. bcurl gives up after `-t` seconds (30) without a RESPONSE or DATA frame of the current request. Exit: 0 all 2xx/3xx; 4 worst 4xx; 5 worst 5xx; 2 connection or protocol error or timeout (later URLs skipped); 1 usage error, including a REQUEST too big for one frame (nothing sent). The worst wins: 2, 5, 4, 0.

12. Conformance checklist

- [ ] Both: big-endian; Length ≤ 16384; unknown types skipped by exactly Length bytes; unknown flags ignored; reserved types never sent; table and literal names; DATA through END; `content-length` checked.
- [ ] Client: IDs 1, 2, 3, …; one request at a time; every RESPONSE and DATA ID checked; protocol errors (§7) end the connection.
- [ ] Server: IDs copied; request errors: 400, connection kept; connection errors: 400 on ID 0, GOAWAY `0x01`, close; bodies read through END; §5.1 and §8 path rules; idle close with GOAWAY `0x00`.